PHYS 708 –Assignment 3 (Due date: 31 October 2025)

Problem

(1) If the maximum energy of cosmic-ray protons is determined by the condition that the particle gyro-radius should be less than the size of the acceleration region, calculate the maximum energy of proton for a spherical source of radius $R = 100$ parsec and a magnetic field strength of $B = 1$ micro-gauss. [You can refer to the equation on Slide 20, Lecture 5 and take $\theta = 90°$].

(2) Muons at rest have a mean lifetime (decay time) of 2.2×10^{-6} seconds. Calculate the relativistic lifetime of a muon of kinetic energy 1 GeV produced from cosmic-ray interaction in the Earth's atmosphere. Compare the relativistic lifetime with the time the muon takes to reach the ground if it is produced at a height of 5 km above the ground. Assume that the muon travels vertically downward.

(3) Using the expression for the emissivity of gamma rays from π^0-decay (Slide 13 of Lecture 7), show that the gamma-ray spectrum follows a power-law similar to that of the cosmic-ray protons at very high energies.

(4) If the energy loss rate of a high-energy electron due to synchrotron losses is given by $\frac{dE}{dt} = -aE^2$, where E is the electron energy and a is a positive constant, show that the life time of the electron is proportional to $\frac{1}{E}$.